🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD 6.G.4

Surface Area Using Nets

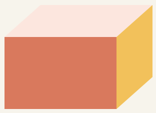
Lesson 10-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

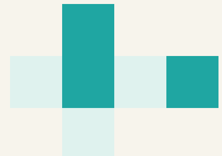


add every face

A box 6 × 4 × 2: SA = 2(24) + 2(12) + 2(8) = 88 in² — like wrapping paper needed

Surface area

Write the definition:



flattened faces

Cut along the edges of a cereal box and unfold it flat — you see all 6 faces

Net

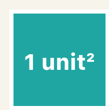
Write the definition:

length × width × height

A rectangular prism has 6 faces: top, bottom, front, back, left, right

Face

Write the definition:

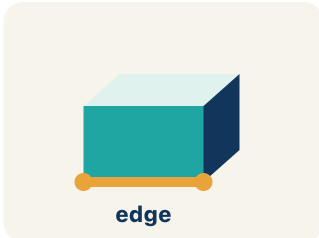


1 by 1 square

A rectangle drawn on paper is 2D — it has length and width but no depth

Two-dimensional

Write the definition:



A cube has 12 edges — the lines where two flat faces connect

Edge

Write the definition:

Guided Notes Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the surface area of a solid by using its net.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Surface area | Net | Face | Two-dimensional | Edge

👆 *Tap any word to see what it means and a picture.*

1. The total area of all the faces of a solid is its ________.
2. A flat pattern that folds up into a solid is a ________.
3. A flat surface of a solid figure is a ________.
4. A flat shape with length and width but no thickness is ________.
5. A line segment where two faces meet is an ________.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: A rectangular prism has l = 6 in, w = 4 in, h = 2 in. What is its surface area?

A. 88 in^2

B. 48 in^2

C. 88 in^3

D. 44 in^2

Step 1 SA = 2(6×4) + 2(6×2) + 2(4×2) = 48 + 24 + 16 = 88 in^2.

✅ **Answer:** A. 88 in^2

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: How many faces does a rectangular prism have?

A. 6

B. 4

C. 8

D. 12

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Surface area is measured in which type of unit?

A. Square units (in^2, cm^2, ft^2)

B. Cubic units (in^3, cm^3, ft^3)

C. Linear units (in, cm, ft)

D. No units needed

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Your team unfolds a rectangular time capsule box into a flat net. How many faces (flat sides) will the net show in all?

A. 6

B. 4

C. 5

D. 8

Show your work:

2. A time capsule box measures 2 in × 3 in × 4 in. Using its net, you add up all six faces. What is the total surface area?

A. 52 in^2

B. 24 in^2

C. 26 in^2

D. 48 in^2

Show your work:

Stretch Your Thinking
Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two boxes have the same volume of 60 in^3. Box A is $5 \times 4 \times 3$. Box B is $10 \times 6 \times 1$. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

Sentence starter: Box A: SA = 2(____) + 2(____) + 2(____) = ____ in^2. Box B: SA = 2(____) + 2(____) + 2(____) = ____ in^2. Box ____ uses less wrapping paper because ____.

Show your work:

Reflect — Exit Ticket

A rectangular prism has l = 9 in, w = 5 in, h = 3 in. What is the surface area?

A. 174 in^2

B. 135 in^2

C. 174 in^3

D. 87 in^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Net
3. **Notes 3:** Face
4. **Notes 4:** Two-dimensional
5. **Notes 5:** Edge
6. **We Try (notes):** A. 6 *— A rectangular prism has 6 faces: top, bottom, front, back, left, and right.*
7. **You Try (notes):** A. Square units (in^2, cm^2, ft^2) *— Surface area measures flat surfaces, so it uses square units. Volume uses cubic units.*
8. **Try It 1:** A. 6 *— A rectangular prism has 6 faces — top, bottom, front, back, left, and right — so its net shows 6 faces.*
9. **Try It 2:** A. 52 in^2 *— From the net:* $SA = 2(2\times3) + 2(2\times4) + 2(3\times4) = 12 + 16 + 24 = 52\ in^2$.
10. **Exit Ticket:** A. 174 in^2 *—* $SA = 2(9\times5) + 2(9\times3) + 2(5\times3) = 90 + 54 + 30 = 174\ in^2$. *Surface area uses square units (*in^2*).*